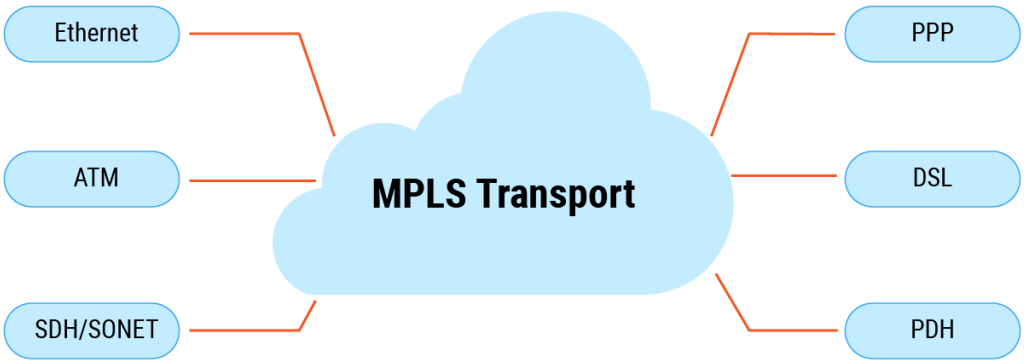
Networks Assignment - 05 Lab Report
Group 07

MPLS Label Forwarding Simulation

1. Objective

The objective of this laboratory experiment is to understand the fundamental operation of **Multiprotocol Label Switching (MPLS)** by simulating the creation of:

- Routing Information Base (RIB) using Dijkstra's algorithm, and
- Label Forwarding Information Base (LFIB) for label-based packet forwarding.



Through this simulation, we aim to:

1. Represent a fixed 4-router network topology.
2. Compute shortest paths (RIBs) between all routers using Dijkstra's algorithm.
3. Define a Forwarding Equivalence Class (FEC) for packets from R0 to R3.
4. Construct LFIBs for ingress, transit, and egress routers using predefined labels.

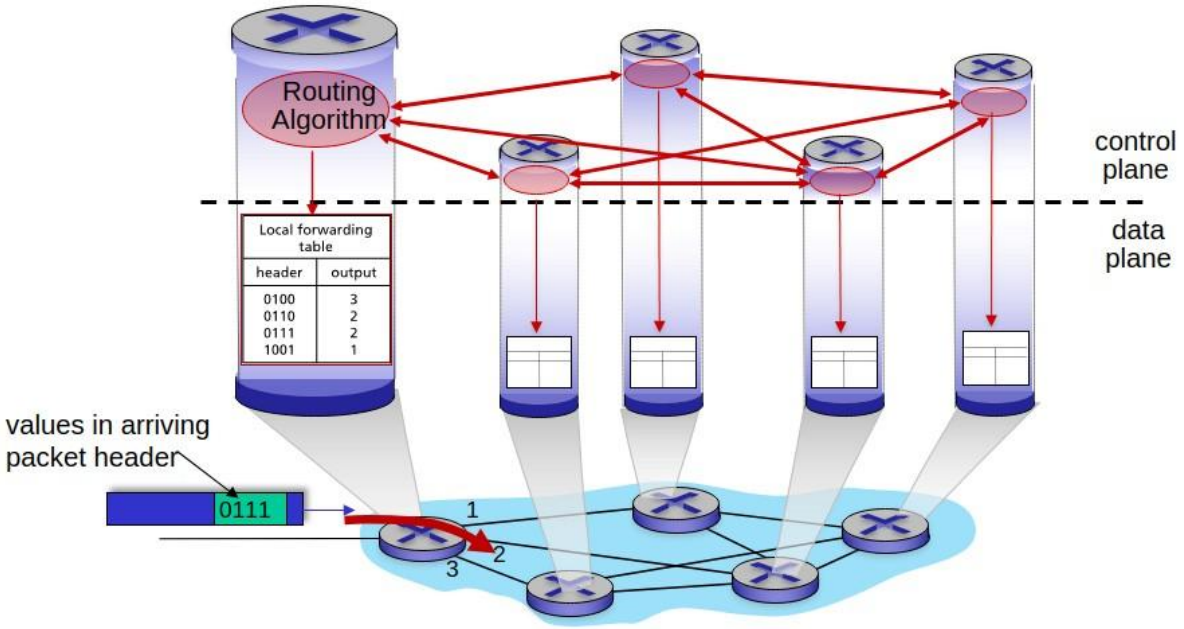
5. Simulate the step-by-step packet forwarding process under MPLS.
6. Compare the MPLS forwarding with normal IP routing.

2. Background Theory

2.1 IP Routing

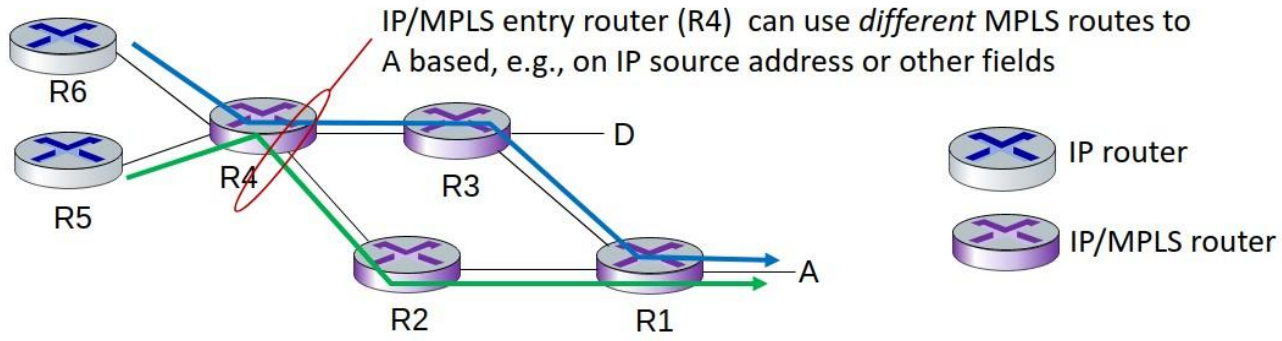
Traditional IP routing uses the destination IP address in each packet to determine the next hop. Every router independently performs a **longest prefix match** lookup in its routing table (RIB).

While flexible, this method becomes computationally expensive as routing tables grow, since every router along the path repeats similar lookups.



2.2 Multiprotocol Label Switching (MPLS)

- **MPLS** is a high-performance forwarding mechanism that replaces complex IP lookups with simple **label-based forwarding**.
- Each packet entering an MPLS network is assigned a short, fixed-length numerical **label**, and routers make forwarding decisions based on that label rather than the IP header.



Key components of MPLS operation:

1. **Ingress Router**: Classifies incoming packets into **Forwarding Equivalence Classes (FECs)** and pushes a label onto the packet.
2. **Transit Router(s)**: Use the incoming label to perform a fast **label swap** operation, replacing it with a new outgoing label.
3. **Egress Router**: Pops (removes) the label and forwards the original IP packet to its destination.

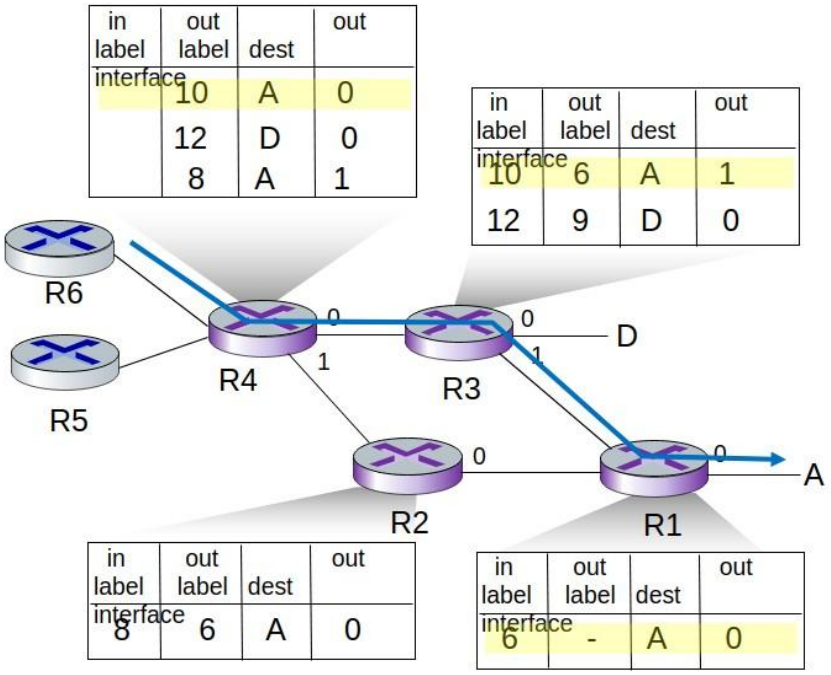
- This label swapping forms a **Label Switched Path (LSP)** — a pre-computed, unidirectional route through the MPLS domain.
- The forwarding tables that store this mapping are known as the **Label Forwarding Information Base (LFIB)**.

2.3 Forwarding Equivalence Class (FEC)

A FEC groups packets that are forwarded the same way through the network — for example, packets from R0 destined to R3. Each router along the path maintains LFIB entries corresponding to the FEC.

2.4 Label Distribution

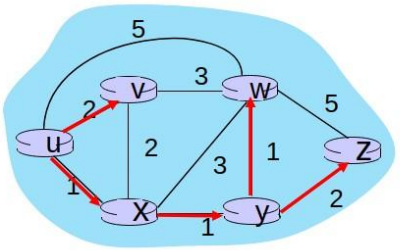
In a real MPLS network, label assignment and distribution are automated via the **Label Distribution Protocol (LDP)** or **RSVP-TE**. However, in this simulation, label values are statically assigned for simplicity.



2.5 Dijkstra's Algorithm

- Dijkstra's algorithm computes the **shortest-path tree** from a source node to all other nodes in a weighted graph.
- It maintains a priority queue of nodes based on minimum cumulative cost and iteratively updates distances to all neighbors.
- This algorithm is used to compute each router's **RIB** in our network.

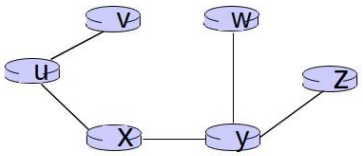
Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					



Initialization (step 0): For all a: if *a* adjacent to then $D(a) = c_{u,a}$

find *a* not in *N'* such that *D(a)* is a minimum
add *a* to *N'*
update *D(b)* for all *b* adjacent to *a* and not in *N'* :
D(b) = min (D(b), D(a) + $c_{a,b}$)

resulting least-cost-path tree from u:

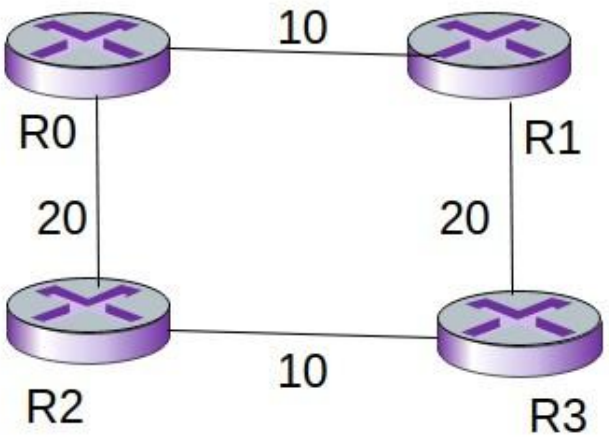


resulting forwarding table in u:

destination	outgoing link	
v	(u,v)	route from *u* to *v* directly
x	(u,x)	route from u to all other destinations via *x*
y	(u,x)	
w	(u,x)	
x	(u,x)	

3. Topology Description

The fixed topology consists of **four routers: R0, R1, R2, and R3** connected as follows:



The goal is to find the shortest paths and then simulate MPLS forwarding for traffic from **R0 (Ingress)** to **R3 (Egress)**.

4. Implementation

4.1 Tools and Language

The simulation was implemented in **C++17**, using STL data structures (vector, priority_queue, and map).

4.2 Task-wise Implementation Summary

Task 1 – Representing the Topology:

The topology is modeled as an **adjacency list**, where each router maintains a list of edges with associated costs.

Task 2 – Computing Routing Tables (RIB):

Each router runs Dijkstra's algorithm independently to compute its **Routing Information Base (RIB)**.

The output table contains:

- Destination router
- Next hop
- Total path cost

Task 3 – Defining FEC:

FEC = All packets originating at **R0** and destined for **R3**.

Task 4 – Creating Label Forwarding Information Bases (LFIBs):

Router	LFIB Entry	Operation
R0 (Ingress)	(FEC: R0→R3) → (Push 300, Send to R2)	Push
R2 (Transit)	(In-Label: 300) → (Out-Label: 777, Send to R3)	Swap
R3 (Egress)	(In-Label: 777) → (Pop Label, Deliver)	Pop

Task 5 – Packet Simulation:

A **Packet** structure was defined with fields:

```
struct Packet {
   int src;     int dest;
optional<int> label;
};
```

The forwarding process follows:

1. R0 pushes label 300 and sends the packet to R2.
2. R2 swaps label 300 → 777 and forwards to R3.
3. R3 pops label 777 and delivers the packet.

5. Results and Analysis

5.1 Routing Tables

Each router's RIB was printed as shown in the output. The computed **shortest path from R0 to R3** was:

R0 → R2 → R3

Total Cost = 30

This matches the theoretical minimum (R0–R2 cost 20 + R2–R3 cost 10 = 30).

5.2 Label Forwarding Behavior

- **Ingress (R0):** Introduced MPLS label (300) into the packet header.
- **Transit (R2):** Swapped 300 → 777 using LFIB mapping.
- **Egress (R3):** Recognized label 777, popped it, and delivered the packet.

No IP lookup was performed inside the MPLS domain (R2), demonstrating the performance advantage of label switching.

5.3 Comparison with Standard IP Routing

Feature	IP Routing	MPLS Forwarding
Forwarding Decision	Based on IP address (RIB)	Based on label (LFIB)
Processing Speed	Slower due to prefix matching	Faster, constant-time label lookup
Routing Table Lookup	At each hop	Only once at ingress

Thus, MPLS optimizes forwarding performance by reducing per-hop processing complexity.

5.4 Output

```
Routing Table for Router R0:
-------------------------------------
|Destination| Next Hop | Total Cost |
-------------------------------------
|    R1     |   R1     |   10       |
|    R2     |   R2     |   20       |
|    R3     |   R1     |   30       |
-------------------------------------

Routing Table for Router R1:
-------------------------------------
|Destination| Next Hop | Total Cost |
-------------------------------------
|    R0     |   R0     |   10       |
|    R2     |   R0     |   30       |
|    R3     |   R3     |   20       |
-------------------------------------
```

```
Routing Table for Router R2:
-------------------------------------
|Destination| Next Hop | Total Cost |
-------------------------------------
|    R0     |   R0     |   20       |
|    R1     |   R3     |   30       |
|    R3     |   R3     |   10       |
-------------------------------------

Routing Table for Router R3:
-------------------------------------
|Destination| Next Hop | Total Cost |
-------------------------------------
|    R0     |   R2     |   30       |
|    R1     |   R1     |   20       |
|    R2     |   R2     |   10       |
-------------------------------------
```

```
Enter source router ID (0-3): 0
Enter destination router ID (0-3): 3
Enable MPLS? (y/n): n

Simulating packet from R0 to R3 using standard IP routing...
[IP Routing] Path: R0 -> R1 -> R3
Packet delivered successfully via standard IP routing.
```

```
Enter source router ID (0-3): 0
Enter destination router ID (0-3): 3
Enable MPLS? (y/n): y

Simulating packet from R0 to R3 using MPLS...
[R0] Packet for R3 (FEC: R0->R3). Pushing Label 300. Sending to R2.
[R2] Received packet with In-Label 300. Swapping for Out-Label 777. Sending to R3.
[R3] Received packet with In-Label 777. Popping label. Packet delivered.
```

6. Discussion

1. **Scalability:**
   MPLS reduces dependency on large IP routing tables and simplifies lookup operations, making it scalable for large backbone networks.

2. **Traffic Engineering:**
   MPLS allows explicit control of LSPs, enabling optimized bandwidth usage and load balancing.

3. **QoS and VPNs:**
   MPLS supports traffic classification and differentiated QoS, forming the basis for MPLS-VPN deployments in modern ISPs.

4. **Simulation Limitations:**
   - Label assignment is static and not dynamically distributed (no LDP simulation).
   - Only a single FEC (R0→R3) is implemented.
   - No failure recovery or loop prevention logic is modeled.

5. **Extensions:**
   Future versions could implement multiple FECs, simulate label distribution protocols, or dynamically recompute paths under link failures.

7. Conclusion

- The experiment successfully demonstrated the end-to-end MPLS forwarding process.
- Starting with the generation of routing tables using Dijkstra's algorithm, we defined a specific FEC, constructed LFIB entries for all routers, and simulated label push, swap, and pop operations.
- The results confirm that **MPLS forwarding achieves the same routing outcome as IP routing**, but with significantly reduced per-hop computation.

- This experiment thus provided a clear conceptual and practical understanding of MPLS label switching and its advantages in modern networking.